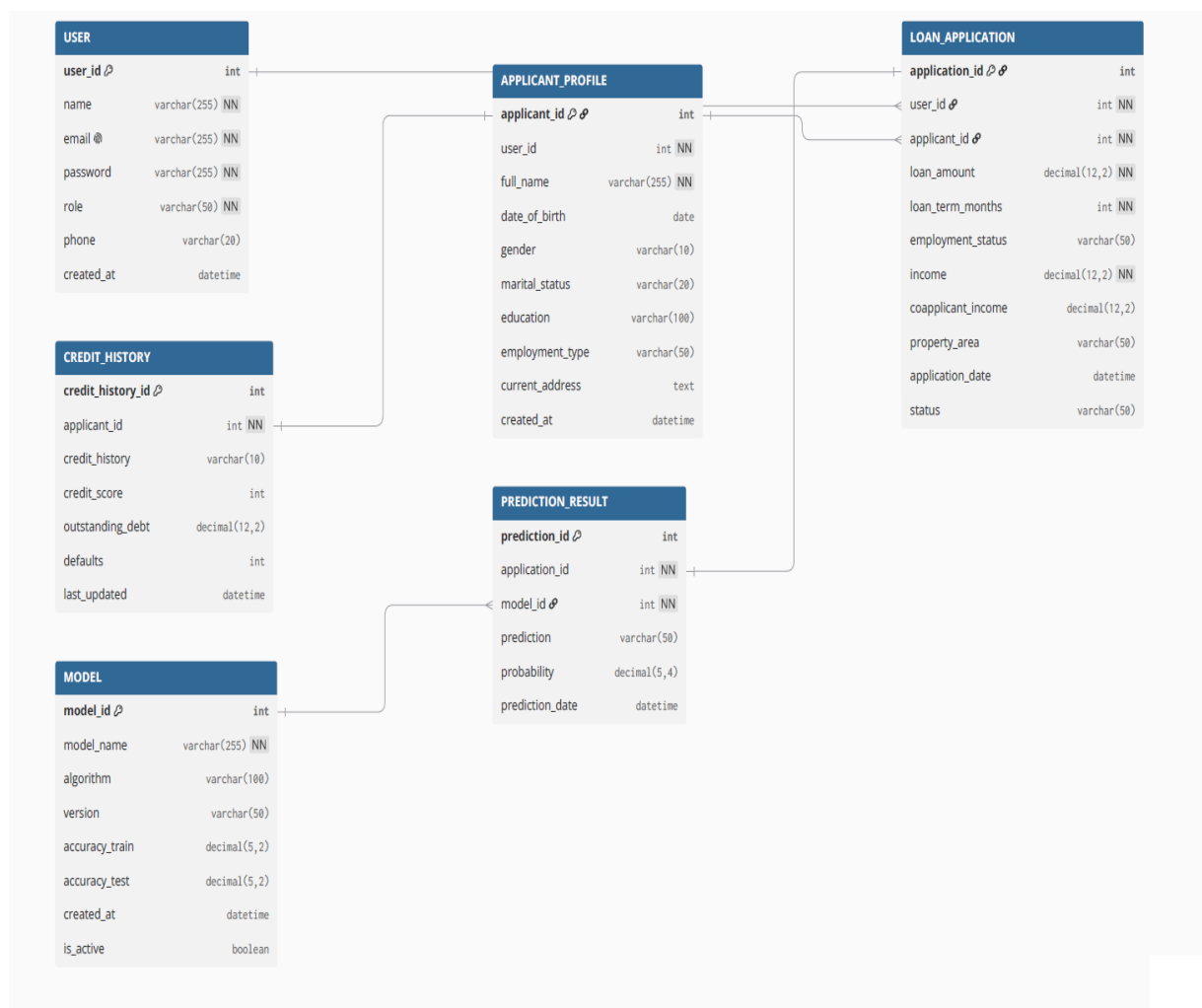


Entity Relationship Diagram (ERD) Specification

Date	28 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender

Entity Relationship Diagram (ERD)



Database Architecture (Schema Specification):

The Smart Lender application maps out data management across 6 core structured entities

Table Name	Primary Key	Key Fields	Description & Purpose
USER	user_id	name, email, password, role, phone	Manages secure login credentials and access authentication.
APPLICANT_PROFILE	applicant_id	user_id(FK), demographics	Store personal applicant demographics linked to a unique user.
CREDIT_HISTORY	credit_history_id	applicant_id(FK), flags	Tracks financial risk profiles, background checks, and outstanding debts.
LOAN_APPLICATION	application_id	applicant_id(FK), income metrics, user_id(FK)	Evaluates individual/co-applicant income and requested amount.
MODEL	model_id	model_name, train/test metrics	Manages internal ML pipeline reference(XG Boost ,KNN , Random Forest).
PREDICTION_RESULT	prediction_id	application_id(FK), model_id(FK)	Connects with application feeds to display approval/rejection outputs.

System Workflow:

- **User Authentication** :The client logs in via the secure USER credentials.
- **Profile & Financial Check** : The system retrieves the APPLICANT_PROFILE and checks creditworthiness flags in CREDIT_HISTORY.
- **Application Intake** : The user submits financial variables through the LOAN_APPLICATION interface.
- **Machine Learning Inference** : The data passes into the trained MODEL pipeline (e.g., XG Boost).
- **Decision Output** : The final approval or rejection status is saved in PREDICTION_RESULT and displayed back to the user.